

A 4D Non-Associative Loop Algebra for Extraneous Solutions

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Abstract

This paper presents the formal algebraic construction of a 4-dimensional, power-associative, globally non-associative algebra over $\mathbb{R}$ spanned by the basis $\{1, i, t, it\}$. This system introduces a specialized radical unit defined by $\sqrt{t} = -1$, (t is currently a placeholder for this unit), designed to capture and evaluate multi-valued extraneous branches that standard associative fields structurally discard. We demonstrate that the multiplicative structure forms a closed Loop (non-associative magma), providing an analytic path-dependent framework viable as a cryptographic primitive for post-quantum key exchange networks.

1 Introduction and Split-Complex Number System Limitations

In standard complex and split-complex fields governed by global associativity, attempting to evaluate the negative root state of a radical unit forces a logical collapse. If an element j satisfies $j^2 = 1$, enforcing the radical condition $\sqrt{j} = -1$ immediately causes:

$$j = (\sqrt{j})^2 = (-1)^2 = 1 \quad (1)$$

This violates the foundational split-complex constraint $j \neq \pm 1$. Standard calculus resolves this by discarding extraneous radical roots. This framework proposes an alternative architecture by abandoning global associativity to preserve path history.

2 Foundational Axioms and Loop Structure

We define a 4-dimensional vector space spanned by the linearly independent basis $\{1, i, t, it\}$ subject to the following three axioms:

1. $t^2 = 1$

2. $t \neq \pm 1$
3. $t^{1/2} = -1$

The multiplicative operations are defined completely by the following non-associative table:

$\cdot$	1	i	t	it
1	1	i	t	it
i	i	-1	it	t
t	t	-it	1	-i
it	it	-t	i	-1

Evaluating the expression $i \cdot i \cdot t$ demonstrates the broken symmetry:

$$(i \cdot i) \cdot t = (-1) \cdot t = -t \quad (2)$$

$$i \cdot (i \cdot t) = i \cdot it = t \quad (3)$$

Since $-t \neq t$, global associativity is destroyed, protecting the definition of $\sqrt{t} = -1$.

For this reason, it is not a value of the imaginary unit i multiplied by t but a unit.

3 Analytic Consistency and Taylor Expansion

Applying the definition of continuous exponents $t^n = e^{n \ln(t)}$, the system yields the precise constraint $\ln(t) = 2\pi i$. Evaluating this via an infinite Taylor series confirms self-consistency:

$$t^n = \sum_{k=0}^{\infty} \frac{(2\pi i n)^k}{k!} = \cos(2\pi n) + i \sin(2\pi n) \quad (4)$$

For $n = 1/2$, $t^{1/2} = \cos(\pi) + i \sin(\pi) = -1$. For $n = 2$, $t^2 = \cos(4\pi) + i \sin(4\pi) = 1$. This demonstrates how while there is not global associativity, local power associativity is preserved as index laws of the single unit t still work. Because multiplication is not associative, we can use this Taylor series expansion to define powers of t .